

One-Dimensional Arrays (Vectors): Traversing Vectors

Translated from Romanian: *Tablouri unidimensionale (vectori) — Parcurgerea vectorilor* (Grade 9)

SOURCE URL pblinfo.ro/probleme/categorii/9/tablouri-unidimensionale-vectori-parcurgerea-vectorilor	GRADE & DIFFICULTY 9th Grade • Easy (Ușoară)	I/O MODE Standard Console (stdin / stdout)	AUTHOR Silviu Candale
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This document provides the complete English translation of the problems listed on the **Parcurgerea vectorilor** (Traversing Vectors) category page on pblinfo.ro. Each entry includes both the translated summary from the category listing and the full problem specification (Requirement, Input/Output Data, Restrictions, and Examples).

#ID	ORIGINAL TITLE	ENGLISH TITLE	CORE OBJECTIVE
#633	Paritate1	Parity1	Absolute difference between the count of even and odd elements in an array.
#546	Afisare0	Display0	Display all elements that are multiples of the last element in the array.
#4383	Inlocuire6	Replace6	Replace the last prime element in the array with 0 and print the array.
#4382	Inlocuire5	Replace5	Replace every prime element in the array with 0 and print the array.
#488	Afisare	Display	Print even-indexed elements in ascending index order, then odd-indexed in descending order.
#489	Afisare1	Display1	Print elements alternating from both ends: first, last, second, second-to-last, etc.
#486	MinMax0	MinMax0	Determine the minimum and maximum values among the elements of an array.
#553	PozMinMax	PosMinMax	Determine the 1-based indices of the minimum and maximum elements in an array.
#547	Numarare6	Counting6	Count how many elements equal the difference between the maximum and minimum values.
#490	AfisareMinMax	DisplayMinMax	Display all elements located between the minimum and maximum elements (inclusive).

1. Paritate1 (Parity1)

#633

Easy

Console I/O

Original Romanian Summary: *Se dă un șir cu n elemente, numere naturale. Determinați diferența în valoare absolută dintre numărul de valori pare din șir și numărul de valori impare din șir.*

Requirement

Given a sequence of n elements (natural numbers), determine the absolute difference between the number of even values in the sequence and the number of odd values in the sequence.

Input Data

The program reads from the keyboard the natural number n , followed by n natural numbers representing the elements of the sequence.

Output Data

The program displays on the screen the number C , representing the required absolute difference.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The n numbers read will be strictly less than $1,000,000,000$.

Input	Output
5 15 245 28 33 11	3

Explanation: There are 4 odd numbers (15, 245, 33, 11) and 1 even number (28) in the sequence, so the absolute difference is $|1 - 4| = 3$.

2. Afisare0 (Display0)

#546 • Py

Easy

Console I/O

Original Romanian Summary: Se citește un vector cu n elemente, numere naturale. Să se afișeze elementele din vector care sunt multipli ai ultimului element.

Requirement

A one-dimensional array (vector) with n elements, natural numbers, is read. Display all elements from the array that are multiples of the last element.

Input Data

The program reads from the keyboard the number n , and then n natural numbers representing the elements of the array.

Output Data

The program displays on the screen the required elements, separated by a single space, in the order they appear in the array.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array will be less than $1,000,000$.
- The last element of the array is guaranteed to be non-zero.

Input	Output
5 7 4 9 6 2	4 6 2

Explanation: The last element is 2. The elements in the array divisible by 2 are 4, 6, and 2.

3. Inlocuire6 (Replace6)

#4383 • Py

Easy

Console I/O

Original Romanian Summary: Se citește un vector cu n elemente, numere naturale. Să se înlocuiască ultimul element prim din vector cu 0, apoi să se afișeze vectorul.

Requirement

A one-dimensional array (vector) with n elements, natural numbers, is read. Replace the last prime element in the array with 0 , then display the updated array.

Input Data

The program reads from the keyboard the number n , and then n natural numbers representing the elements of the array.

Output Data

The program displays on the screen the elements of the array after the replacement, separated by single spaces.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array will be less than $1,000,000$.

Input	Output
5 7 4 9 6 2	7 4 9 6 0

Explanation: The prime numbers in the array are 7 and 2. The last prime number is 2, which is replaced with 0.

4. Inlocuire5 (Replace5)

#4382

Easy

Console I/O

Original Romanian Summary: Se citește un vector cu n elemente, numere naturale. Să se înlocuiască fiecare element prim din vector cu 0, apoi să se afișeze vectorul.

Requirement

A one-dimensional array (vector) with n elements, natural numbers, is read. Replace every prime element in the array with 0 , then display the updated array.

Input Data

The program reads from the keyboard the number n , and then n natural numbers representing the elements of the array.

Output Data

The program displays on the screen the elements of the array after the replacement, separated by single spaces.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array will be less than $1,000,000$.

Input	Output
5 7 4 9 6 2	0 4 9 6 0

Explanation: Both prime elements (7 and 2) are replaced with 0.

5. Afisare (Display)

#488

Easy

Console I/O

Original Romanian Summary: Se citește un vector cu n elemente, numere naturale. Să se afișeze elementele cu indici pari în ordinea crescătoare a indicilor, iar elementele cu indici impari în ordinea descrescătoare a indicilor.

Requirement

A one-dimensional array (vector) with n elements, natural numbers, is read. Display the elements at even indices in ascending order of their indices, and the elements at odd indices in descending order of their indices.

Input Data

The program reads from the keyboard the number n , and then n natural numbers representing the elements of the array.

Output Data

The program displays on the first line of the screen the elements with even indices, in ascending order of indices, separated by a single space, and on the second line the elements with odd indices, in descending order of indices, separated by a single space.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array will be less than **1,000**.
- Index numbering starts from **1** (1-based indexing).

Input	Output
5 7 9 2 6 8	9 6 8 2 7

Explanation: Elements at even positions (indices **2, 4**) are **9, 6**. Elements at odd positions in reverse order (indices **5, 3, 1**) are **8, 2, 7**.

6. Afisare1 (Display1)

#489

Easy

Console I/O

Original Romanian Summary: Se citește un vector cu n elemente, numere naturale. Să se afișeze elementele vectorului în următoarea ordine: primul, ultimul, al doilea, penultimul, etc.

Requirement

A one-dimensional array (vector) with n elements, natural numbers, is read. Display the elements of the array in the following alternating order: first, last, second, second-to-last, third, third-to-last, etc.

Input Data

The program reads from the keyboard the number n , and then n natural numbers representing the elements of the array.

Output Data

The program displays on the screen, separated by a single space, the elements of the array in the required order.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array will be less than **1,000**.

Input	Output
5 2 9 1 5 8	2 8 9 5 1

Explanation: First (2), last (8), second (9), second-to-last (5), and middle (1).

7. MinMax0 (MinMax0)

#486 • Py

Easy

Console I/O

Original Romanian Summary: *Să se determine maximul și minimul valorilor elementelor unui vector.*

Requirement

Read n , and then n natural numbers representing the elements of a one-dimensional array. Calculate the minimum value **minim** and the maximum value **maxim** among the elements of the array.

Input Data

The program reads from the keyboard the number n , followed by n natural numbers representing the elements of the array.

Output Data

The program displays on the screen the two numbers **minim** and **maxim**, separated by a single space.

Restrictions and Clarifications

- $0 < n < 1,000$
- The elements of the array will be less than **1,000,000**.

Input	Output
5 8 2 9 4 5	2 9

Explanation: The array contains 5 elements. The minimum value is 2 and the maximum value is 9.

8. PozMinMax (PosMinMax)

#553

Easy

Console I/O

Original Romanian Summary: *Să se determine indicele maximului și cel al minimului valorilor elementelor unui vector.*

Requirement

Read n , and then n distinct natural numbers representing the elements of a one-dimensional array. Determine the index **imin** of the minimum element and the index **imax** of the maximum element in the given array.

Input Data

The program reads the number n , followed by n natural numbers representing the elements of the array.

Output Data

The program displays on the screen the two indices **imin** and **imax**, separated by a single space.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array are distinct, less than **1,000,000**, and indexed from **1** to n .

Input	Output
5 8 7 9 2 5	4 3

Explanation: The array has 5 elements. The minimum value is 2 at index 4, and the maximum value is 9 at index 3.

9. Numarare6 (Counting6)

#547 • Py

Easy

Console I/O

Original Romanian Summary: Se citește un vector cu n elemente, numere naturale. Să se determine câte elemente ale vectorului sunt egale cu diferența dintre cea mai mare și cea mai mică valoare din vector.

Requirement

A one-dimensional array (vector) with n elements, natural numbers, is read. Determine how many elements of the array are equal to the difference between the largest value and the smallest value in the array.

Input Data

The program reads from the keyboard the number n , and then n natural numbers representing the elements of the array.

Output Data

The program displays on the screen the required count.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array will be less than 1,000,000.

Input	Output
5 7 7 9 2 4	2

Explanation: The difference between the largest and smallest value in the array is $9 - 2 = 7$. The value 7 appears in the array 2 times.

10. AfisareMinMax (DisplayMinMax)

#490 • Py

Easy

Console I/O

Original Romanian Summary: Se citește un vector cu n elemente, numere naturale distincte. Să se afișeze elementele cuprinse între elementul cu valoarea minimă și cel cu valoare maximă din vector, inclusiv acestea.

Requirement

A one-dimensional array (vector) with n elements, distinct natural numbers, is read. Display all elements located between the element with the minimum value and the element with the maximum value in the array, inclusive, in the order they appear in the array.

Input Data

The program reads from the keyboard the number n , and then n distinct natural numbers representing the elements of the array.

Output Data

The program displays on the screen the required elements, separated by a single space.

Restrictions and Clarifications

- $1 \leq n \leq 1,000$
- The elements of the array will be less than **100,000**.

Input	Output
5 7 9 6 2 8	9 6 2

Explanation: The maximum element is **9** (at position 2) and the minimum element is **2** (at position 4). The subarray between them, inclusive, is **9, 6, 2**.